

HOLISTIC BUILD DOCUMENT - AETHER INTELLIGENCE:

Team: AETHER INTELLIGENCE

Build: Intelligent Emergency Response Infrastructure

Institution: University of Limpopo & Mopani TVET

Category: Techno-mania Innovation Challenge

CONTEXT & PROBLEM STATEMENT

1.CONTEXT

We are building an intelligent emergency-response platform to improve how emergency incidents, patients, emergency vehicles, healthcare facilities and response assets are monitored and coordinated.

The system combines:

- Artificial Intelligence & IoT
- 3D Digital Twin technology
- Smart Real-time telemetry

The platform is designed around a simple operational principle:

Sense → Locate → Understand → Decide → Allocate → Respond → Track

2.THE PROBLEM STATEMENT

Emergency response is fundamentally a **time-sensitive coordination problem**. When an individual experiences a medical emergency, the system needs to answer several questions immediately:

- **Where is the patient?**
- **What is happening?**
- **How severe the situation be?**
- **Which emergency resources are available?**
- **Where are those resources?**

These questions involve information from different physical and digital systems.

The challenge is therefore: **How can emergency authorities convert fragmented real-world information into a unified, real-time operational picture that supports faster and better decisions?**

In Limpopo, an emergency can become life-threatening not because help does not exist, but because the right help may not know where it is, where it is needed, or how to reach the person in time. Consider a person living alone in a deep-rural community. It could be an elderly person, a 30-year-old. The problem becomes even more severe in Limpopo's mountainous and geographically dispersed communities, where network coverage remains unreliable. A person may need emergency assistance but be unable to make a call, send their location, or communicate with responders due to network constraints. This is not only a problem of insufficient resources. It is also a problem of resource

visibility, coordination and utilisation. For a province as geographically complex as Limpopo, this creates a critical question: **What if emergency services could see the province as a live, continuously changing system— knowing where people need help, where ambulances and responders are, which facilities have capacity, and which available resource can reach an incident fastest?** The challenge is therefore not simply to acquire more emergency resources. The challenge is to make the resources that already exist more visible, intelligent, coordinated and responsive—particularly for people living alone and communities where distance, terrain and network limitations can turn minutes into life-or-death consequences

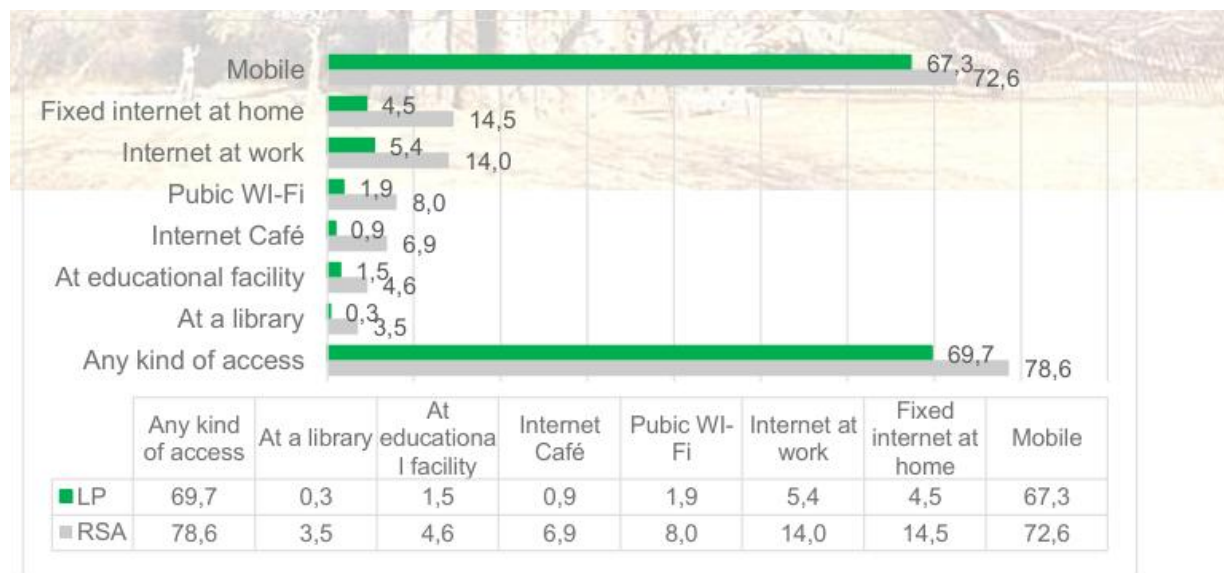


Figure 38: % distribution of households with access to the Internet in Limpopo and RSA

Why this problem matters (Validation)

Limpopo's geography includes large rural areas and geographically dispersed communities. Emergency resources must therefore operate across substantial distances.

More recently, the province has explicitly prioritised improving emergency medical response through *better ambulance coordination and deployment in high-demand and rural areas*. ([Limpopo-Health-sets-bold-target-to-improve-access-and-quality-health-care-in-202627.pdf](#))

And in 2025, Limpopo announced a provincial *Computer Aided Dispatch* system intended to improve ambulance response speed and accuracy. ([Limpopo-Health-sets-bold-target-to-improve-access-and-quality-health-care-in-202627.pdf](#)).

Emergency response does not only need more vehicles. It needs better intelligence about where the emergency is, what is happening, which resources are available, and how those resources should be coordinated. SKYNET creates that intelligence layer.

Limpopo's Department of Health said in 2026, it is prioritising stronger emergency medical response times through **better coordination and deployment of ambulances, particularly in high-demand and rural areas**. ([Limpopo-Health-sets-bold-target-to-improve-access-and-quality-health-care-in-202627.pdf](#))

PROBLEM VALIDATION

Institutional Validation - The problem is consistent with current provincial priorities.

Limpopo Health's 2026/27 priorities explicitly include:

- strengthening emergency medical response times
- better coordination and deployment of ambulances
- addressing high-demand and rural areas ([Limpopo-Health-sets-bold-target-to-improve-access-and-quality-health-care-in-202627.pdf](#))

That gives us **external validation that the problem is real and strategically relevant.**

Technology/Policy Validation - South Africa's National Digital Health Strategy explicitly recognises the role of IoT, AI and smart devices in the evolution of digital health and identifies the importance of:

- integrated digital platforms
- mobile health
- digital health applications
- network infrastructure ([national-digital-strategy-for-south-africa-2019-2024-b.pdf](#))
(<https://www.health.gov.za/wp-content/uploads/2025/07/Web-Version-The-Emerging-South-African-Digital-Health-System.pdf>)

Our solution is designed around the same broad direction as South Africa's digital-health transformation: interconnected information systems, mobile health technologies, interoperable data and digitally enabled service delivery.

Stakeholder Consultation

Stakeholder consulted	Role	Problem discussed	Major feedback
Mankweng Police Service	Information and Response Staff	Difficulty coordinating resources and information gathering	Emphasised the importance of knowing where incidents are occurring, where response vehicles are located.
Mankweng Department of Health	Emergency medical response and ambulance coordination	Difficulty coordinating available ambulances and challenges reaching rural areas.	Highlighted the need for better ambulance coordination, visibility of available resources

The stakeholder discussions helped us move from a broad problem of “**poor emergency response**” to a more specific and technically addressable problem:

Emergency-response institutions need better real-time situational awareness and coordination of geographically distributed resources so that the nearest available emergency resources can be identified and deployed more efficiently, particularly in high-demand and difficult-to-reach areas.

SOLUTION OVERVIEW

A REAL-TIME EMERGENCY INTELLIGENCE PLATFORM

Our solution involves a real-time emergency intelligence infrastructure designed to help with **intelligent, coordinated response**.

Instead of relying primarily on fragmented information, phone calls and estimates, we create **a live digital twin of the environment**—giving decision-makers a continuously updated view of incidents, patients and other response resources.

The platform **connects people, resources and intelligent decision-making through one system**.

1 THE DIGITAL TWIN

At the centre is a real-time 3D digital twin. It provides a visual representation of the environment and continuously maps relevant emergency information. The digital twin is the **operational representation of the emergency environment**.

It contains:

Static information

- Roads and Buildings
- Hospitals and Clinics

Dynamic information

- Ambulances
- Traffic
- Wearable telemetry

Intelligence

- Incident priority
- Resource availability

- Routing

This allows coordinators to move beyond asking: **"Where is the nearest ambulance?"** and instead ask: **"Which available resource can respond to this incident most effectively, based on its current location, status, route and the severity of the emergency?"**. The system therefore support **real-time resource allocation and coordination**.

2 THE SMART IoT WEARABLE

The second component connects the individual directly to the emergency-response system.

The wearable is the **patient-to-system interface: Emergency activation → GPS location → vital-sign telemetry → dashboard**

When the user presses the emergency button, the system transmits the available patient information and location to the platform.

Functions

- Emergency activation
- GPS positioning
- Heart-rate monitoring
- Who requires assistance
- Where the person is located
- Which emergency resources are available nearby

The wearable: **provides physiological telemetry for situational awareness and emergency triage support.**

This becomes particularly important for people living alone, people in remote communities, and individuals who may be unable to communicate their location during an emergency.

Because emergency technology must work for more than one type of user, the device can provide multiple forms of feedback:

- **Sound** for users who can hear
- **Vibration** for users who may not hear an alert
- **Visual** indicators where appropriate

If the emergency button is accidentally activated, the wearable can provide an immediate cancellation mechanism, allowing the user to deactivate a false alarm before unnecessary resources are dispatched.

3 MEDICAL DRONES

In areas where distance, terrain or road accessibility makes conventional response difficult, the platform can coordinate **remotely operated aerial assets**.

The drone provides rapid aerial access and can be used for:

- Emergency assessment
- Delivery of lightweight medical supplies
- Communication support
- Reaching difficult terrain
- live/recorded video;

The human dispatcher remains in control. It provides the real-time intelligence required to make a faster and better-informed decision.



BILL OF MATERIAL SUMMARY

35. BOM SUMMARY

The Holistic Build Document should **not reproduce the entire spreadsheet**.

Instead:

Core physical systems

System	Major Components
Smart Watch	ESP32-S3, display, GPS, heart-rate sensor, battery
Medical Rover	4WD chassis, motors, ESP32, motor drivers, camera, GPS, ultrasonic sensors
Emergency Drone	F450 frame, motors, ESCs, Pixhawk, GPS, transmitter/receiver, battery
Digital Twin	Development workstation, web application, GIS data, database
Connectivity	Wi-Fi/Bluetooth, APIs, telemetry, WebSockets/MQTT

Then:

Detailed component quantities, supplier information, stock codes, prices and purchase links are provided in the accompanying Bill of Materials spreadsheet.

We will link the actual Excel file here once created.

MARKETABILITY AND VALIDATION SECTION

1 Marketability

Aether Intelligence is positioned as an **Emergency Intelligence Infrastructure Platform** that connects real-world emergency data with a real-time digital twin, intelligence and decision-support capabilities.

Primary Customers

Our initial target customers are:

- Provincial Departments of Health
- Hospitals Networks
- Rural Healthcare Network
- Emergency Medical Services (EMS)

Future Markets

The same platform architecture can be adapted to other sectors where real-time spatial intelligence and resource coordination are critical:

- Police and Public Safety
- Mining and Industrial Operations
- Department of Road and Transportation
- Large-Scale Disaster and Humanitarian Response
- National Department of Health

The underlying technology is therefore not limited to one ambulance-dispatch use case.

2 Community Validation

The team conducted informal community-level validation by asking members of the public whether an emergency-intelligence system such as ours would be useful and whether they would be comfortable using such a system themselves or having family members use it.

The responses showed a generally positive perception of the concept.

Key Feedback

A significant number of participants indicated that:

- The system could be useful during emergencies;
- They would be willing to use such a system;
- They would be comfortable with family members using it;
- Faster emergency response would provide meaningful value;
- The system could be particularly useful for people living far from emergency services;

- Rural communities could benefit from improved emergency access.

However, some participants indicated that they would personally not use the system, while still recognising its potential value for vulnerable groups, particularly:

- **elderly people;**
- **people living with epilepsy;**
- people who may be unable to manually request assistance during an emergency.

Validated Problem Statement

The validation process refined the original problem from a general concern about emergency response into a more specific problem:

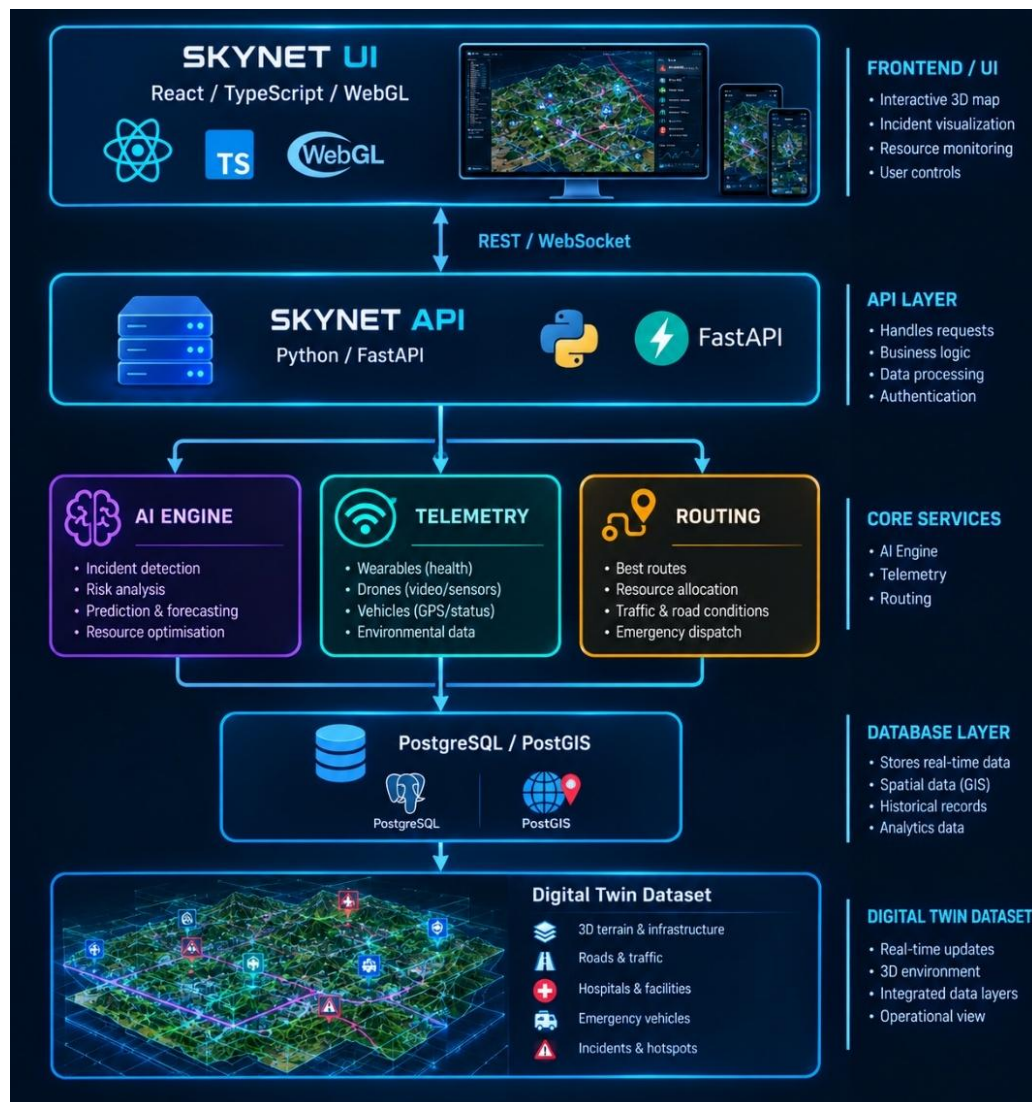
Emergency-response services and communities, particularly in rural and geographically difficult areas, can experience challenges in obtaining timely situational information, coordinating available resources and accessing emergency assistance. Existing communication and information limitations can increase uncertainty during emergencies and make effective resource deployment more difficult.

The objective is not to replace emergency personnel. Instead, the system is designed to **give emergency personnel better information so that they can make faster and more informed decisions.**

TECHNICAL IMPLEMENTATION SECTION

SOFTWARE ARCHITECTURE

The software stack can be represented as:



Frontend - React + TypeScript

Responsible for:

- Digital twin rendering
- Patient information
- Real-time updates

Backend - Python + FastAPI

Responsible for:

- AI integration
- resource allocation
- communication between subsystems

Database - PostgreSQL + PostGIS

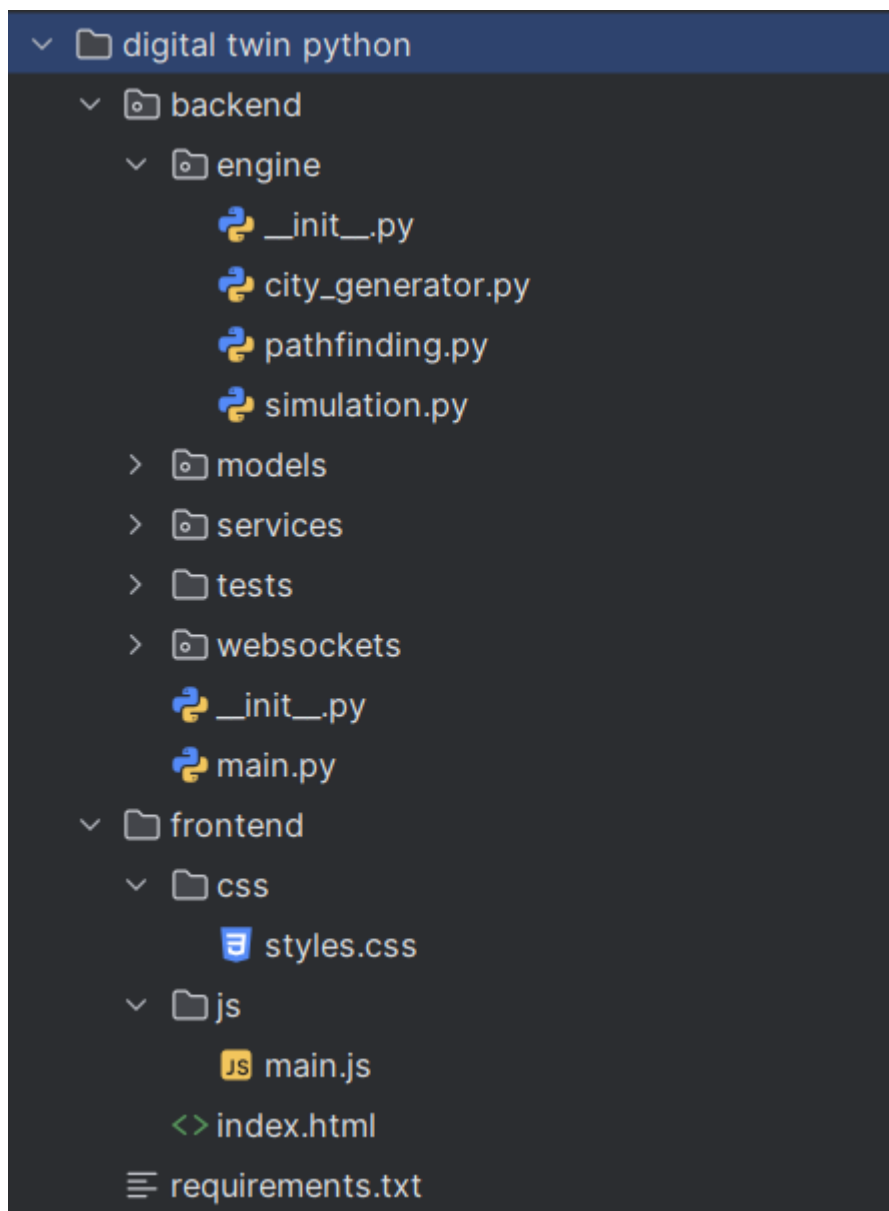
Responsible for:

- coordinates;
- vehicles;
- patients;

Real-time communication - WebSockets / MQTT

Responsible for:

- wearable telemetry;
- drone telemetry;
- live dashboard updates.



MECHANICAL IMPLEMENTATION

The mechanical subsystem consists of:

Watch

- Display housing
- Wrist strap
- Sensor openings
- button/SOS mechanism

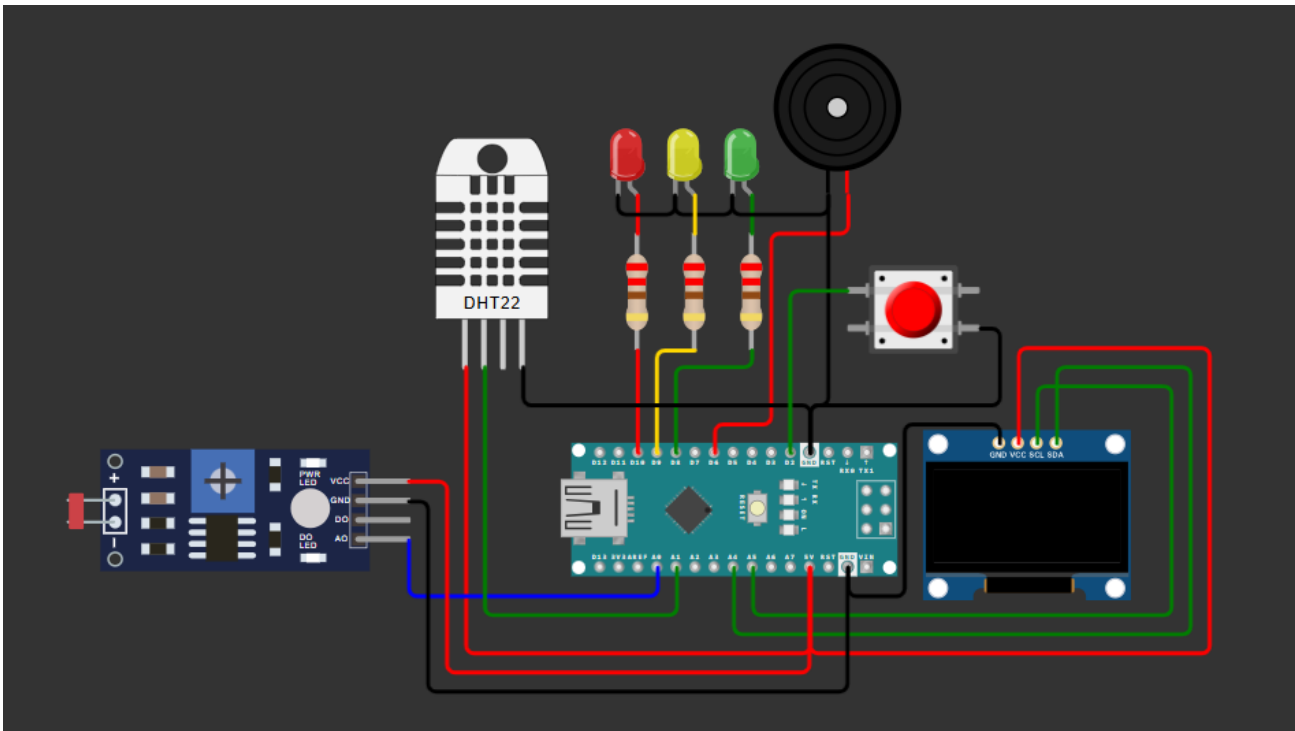


ELECTRONIC IMPLEMENTATION

Watch

Arduino-nano-

- └ Display
- └ Touch
- └ GPS
- └ Heart Rate Sensor
- └ Battery
- └ Wireless



PROGRAMMING IMPLEMENTATION

Watch firmware

- sensor acquisition;
- GPS acquisition;
- SOS event;
- telemetry transmission;
- display interface.

Drone

- flight-controller configuration;
- telemetry;
- GPS;
- mission/remote control.

SKYNET backend

- REST API;
- WebSocket server;
- device registration;
- incident creation;
- telemetry ingestion;

- resource allocation.

SKYNET frontend

- 3D digital twin;
- live asset tracking;
- incident interface;
- dashboard;
- analytics.